Vectors and matrices

This chapter introduces the objects that the rest of the course computes with: vectors, the inner product that measures how two vectors line up, the norm that measures length, and matrices that act on vectors.

Vectors

A vector is an ordered list of numbers. The number of entries is the size of the vector, so a vector with three entries has size 3. We write a vector as a column, for example $x = (2, -1, 2)$, and we write x_i for its i-th entry, so here $x_1 = 2$ and $x_3 = 2$.

Two vectors of the same size can be added by adding matching entries, and a vector can be multiplied by a number by multiplying every entry. For $a = (1, 2, 3)$ and $b = (4, 0, -1)$ the sum is $a + b = (5, 2, 2)$, and twice the first vector is $2a = (2, 4, 6)$. Vectors of different sizes cannot be added.

Inner product

The inner product of two vectors of the same size is the number $a^T b = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$. To compute it, multiply the matching entries and add up the products. The inner product is a single number, not a vector.

As a worked example, take $a = (2, -1, 2)$ and $b = (1, 4, 1)$. Then $a^T b = 2 \cdot 1 + (-1) \cdot 4 + 2 \cdot 1 = 0$. The order does not matter: $a^T b = b^T a$ for every pair of vectors.

Two vectors are called orthogonal when their inner product is zero. The vectors a and b above are orthogonal. A vector whose inner product with itself is zero must be the zero vector.

Norms and distance

The norm of a vector is its length, defined by $\|a\| = \sqrt{a^T a}$. For $a = (2, -1, 2)$ we get $a^T a = 4 + 1 + 4 = 9$, so $\|a\| = 3$. A norm is never negative, and it is zero only for the zero vector.

The distance between two vectors a and b of the same size is the norm of their difference, $\|a - b\|$. Two vectors are close when this number is small.

Scaling a vector scales its norm by the size of the number: $\|ca\| = |c| \, \|a\|$. In particular, multiplying a vector by -1 does not change its norm.

Matrices

A matrix is a rectangular array of numbers. A matrix with m rows and n columns is called an m by n matrix. Two matrices of the same size are added entry by entry, exactly as vectors are.

The product of an m by n matrix A and an n by p matrix B is the m by p matrix $C = AB$ whose entry in row i and column j is the inner product of row i of A with column j of B. The product is only defined when the number of columns of A equals the number of rows of B.

Matrix multiplication is not commutative. In general AB and BA are different matrices, and one of them may not be defined at all. It is associative, however: $(AB)C = A(BC)$.

The transpose of a matrix swaps its rows and columns. The transpose of an m by n matrix is an n by m matrix, and the transpose of a product reverses the order: $(AB)^T = B^T A^T$.